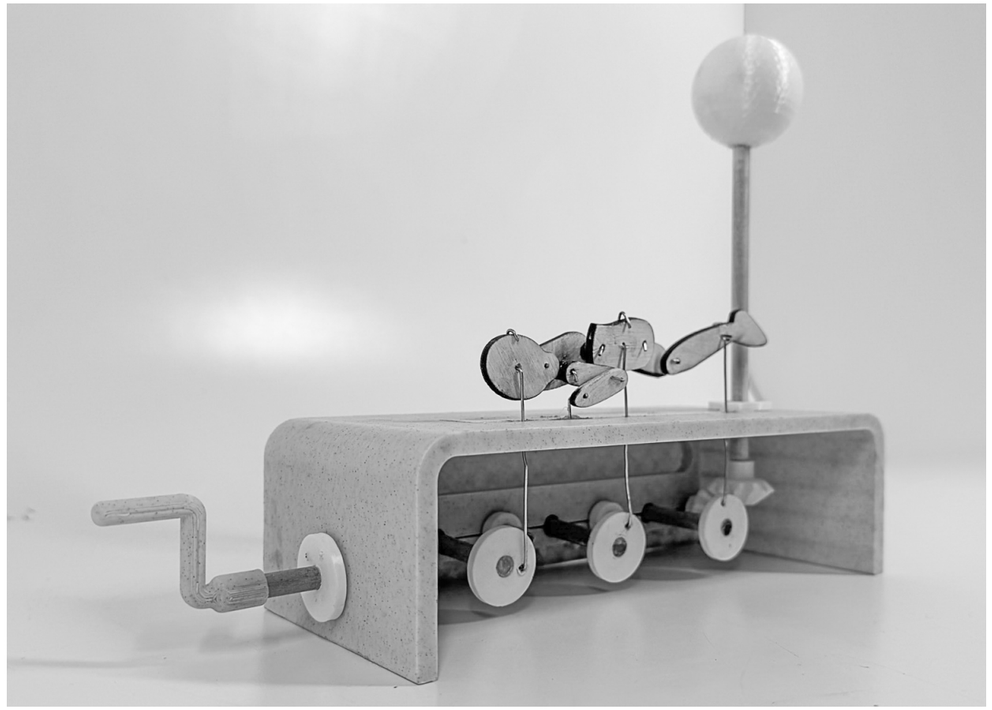
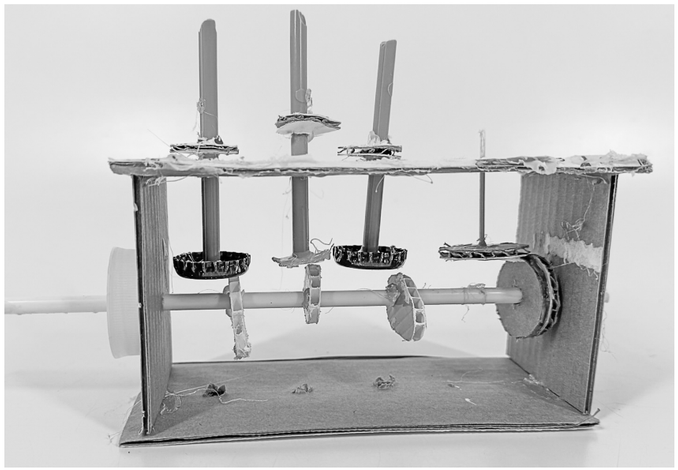
Worm at the Disco

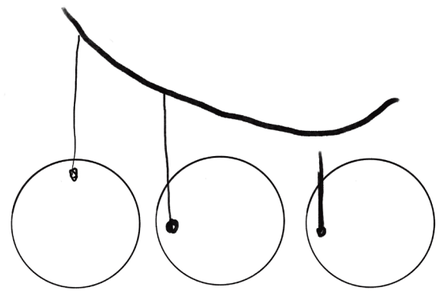


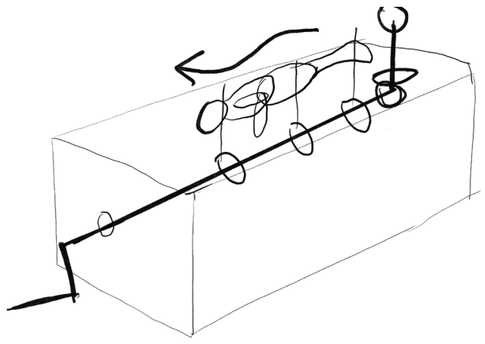
Worm at the Disco is a 3D-printed and laser-cut automaton that blends precise mechanics with playful storytelling. Inspired by cam-driven motion and the humor of the “worm” dance, it transforms simple rotation into an engaging performance. Turning the handle makes the figure dance while the disco ball spins, emphasizing rhythm and movement. This project showcases the expressive potential of mechanical systems in design, merging function with entertainment.

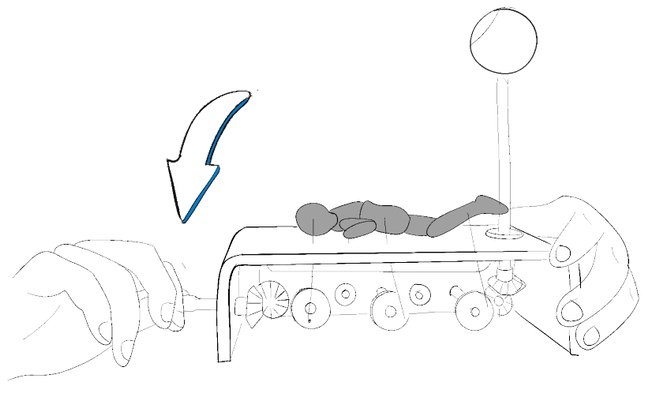


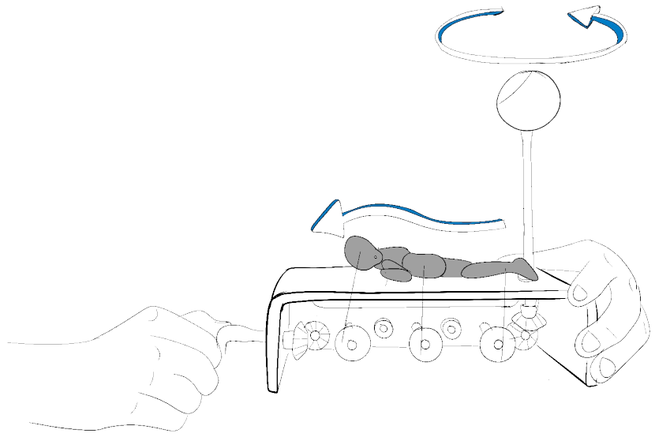


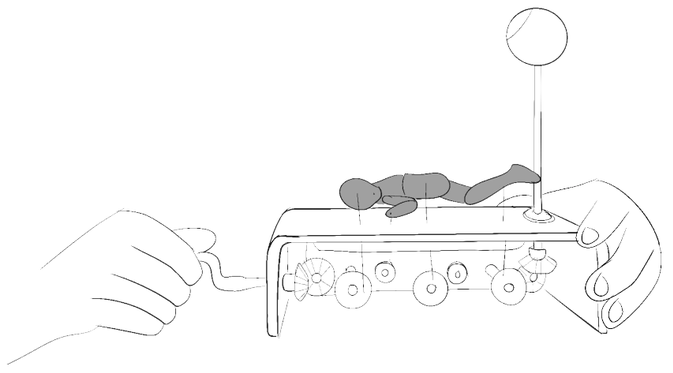
This video demonstrates four key dance movements, translated into precise cam placements for accurate motion. The initial prototype used push rods and lobe cams but proved unreliable. The final design achieved smooth movement using circular cams and gears to drive the axles.

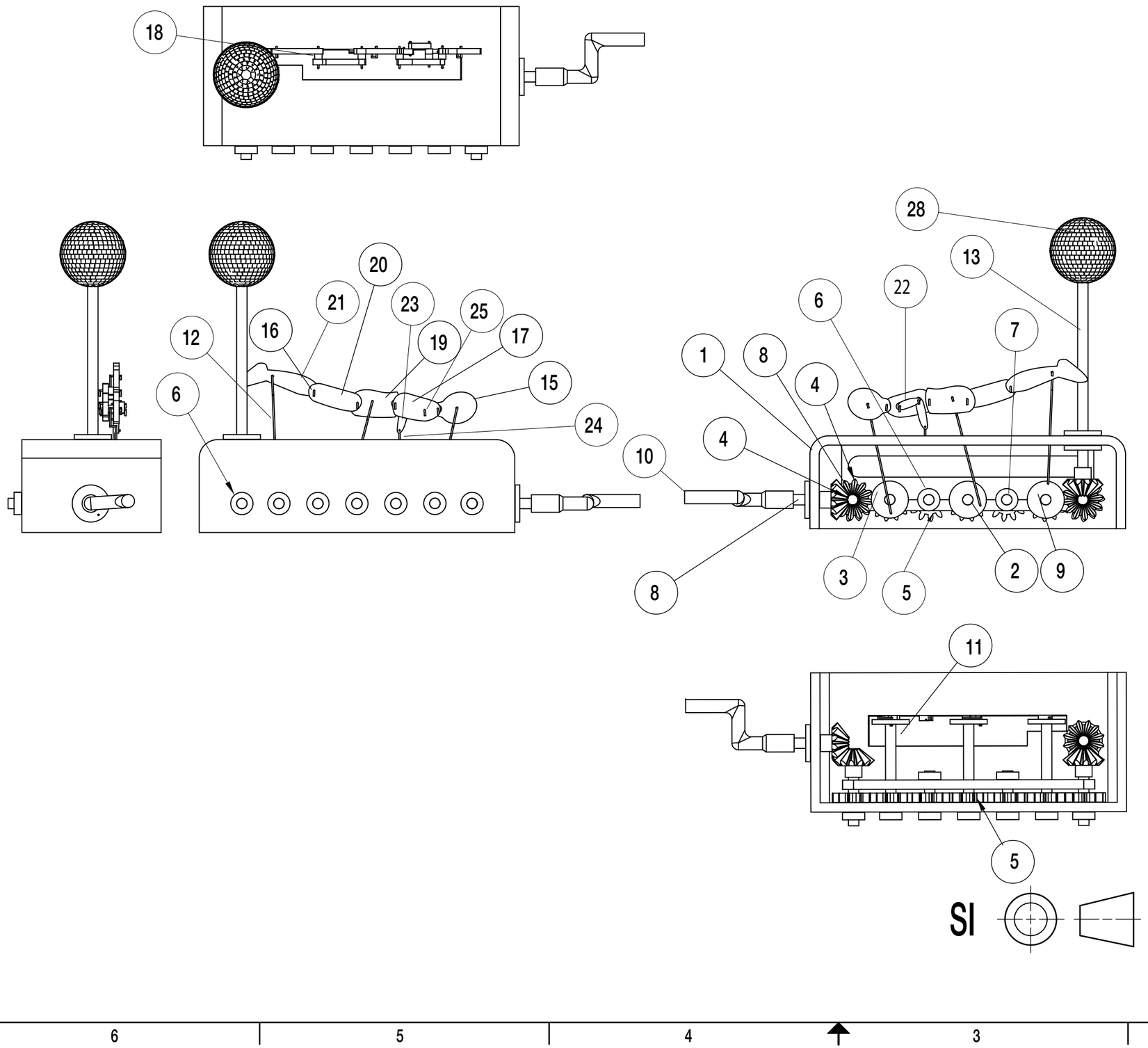












ITEM NO.	PART NUMBER	QTY.
1	Box	1
2	275Shaft	3
3	CircleCam	5
4	Bevel-Base-Revolve	4
5	Inch - Spur gear 12DP 12T 20PA .25FW --- S12N3.0H2.0L0.3125N	7
6	Collar	9
7	Short Peg	2
8	GearShaft	3
9	RearLegsCircleCam	1
10	Crank	1
11	Retainer	1
12	ManWire	3
13	DiscoballShaft	1
14	discoball	1
15	ManHead	1
16	BodyConnector	4
17	ManChest	1
18	Spacer	6
19	ManLower	1
20	ManThigh	1
21	ManLeg	1
22	ManBicep	1
23	ManForearm	1
24	ArmConnector	1
25	ShoulderConnector	1
26	ElbowConnector	1
27	Comb	1
28	Disco Ball	1

CARLETON SCHOOL OF INDUSTRIAL DESIGN
OTTAWA, CANADA

TITLE:	Worm at the Disco		TOLERANCES: DECIMAL ± 0.05 ANGLE ± 0.5 °
DRAWN:	Will Wingrove		
CLIENT:	--		UNITS: IN
DWG NO:	1	REV: --	SCALE: 1:3
DATE:	2025/03/05	SIZE: 11x17	SHEET 1 OF 1